Unit 02: Introduction to Data Mining

CONTENTS

Objectives

After this unit you will be able to:

- Know the concept of data mining
- Understand the knowledge discovery process
- Analyze different types of data repositories
- Understand various data mining functionalities.
- Know the various tasks and current trends in data mining
- Learn various data mining issues

Introduction

Data mining states to the withdrawal of hidden analytical information from huge databases. Data mining techniques can produce the benefits of computerization on prevailing software and hardware stages. Tools of Data mining can respond to business queries that traditional methods were too time-consuming to determine.

2.1 Data Mining

Data mining is the act of consequently looking through huge stores of data to discover patterns and trends that go beyond simple analysis. It utilizes refined mathematical algorithms to segment the data and evaluate the probability of upcoming events. Some key terms to know before going into further detail in Data Mining.

Data

Data are the raw facts, figures, numbers, or text that can be processed by a computer. Today, organizations are gathering massive and growing amounts of data in different formats and different databases.

The operational or transactional data contains the day-to-day operation data (such inventory data, on-line shopping data), non-operational data, and metadata i.e. data about data.

Information

The arrangements, relations, or associations among all types of data can deliver information.

Which products are selling when are based upon the analysis of sales transactions by considering a retail idea.

Knowledge

Information can be converted into knowledge.

Supermarket sales information can be analyzed because of marketing efforts to deliver knowledge of consumer purchasing habits.

Data together in large data repositories develop "data tombs". Data tombs are converted into "golden nuggets" of knowledge with the use of data mining tools see in Figure 1. Golden nuggets mean "small but valuable facts". Data mining is also called as mining of knowledge from data, extraction of knowledge, data/arrangement analysis, data -archaeology, and data-dredging.

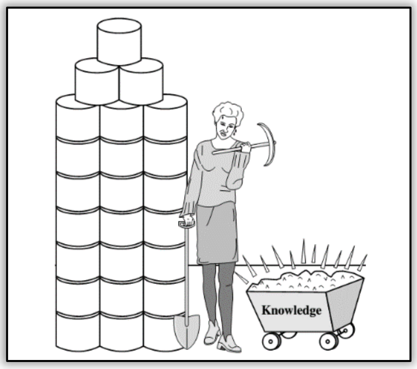

Figure 1: Mining of Data

Why Data Mining?

I am not able to find the data I need (Data is dispersed over the network).The data I have access to is poorly documented (Proper information is missing).I am not able to use the data I have (Unexpected results).Figure 2 shows why data mining is needed.

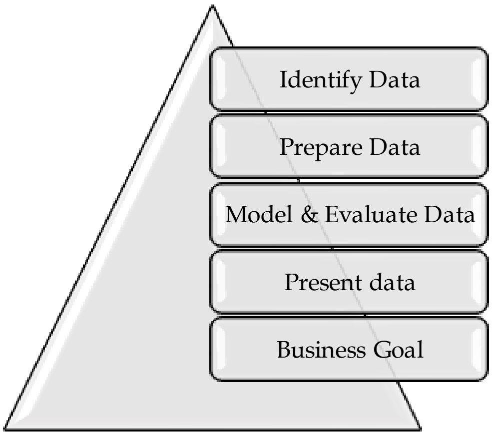

Figure 2: Need for Data Mining

2.2 Process of Knowledge Discovery

Let us have an overview of the steps one by one:

1. Data cleaning: It refers to the removal of inconsistent and noisy data.
2. Integration of data: Multiple information sources are joined during the integration process.

3. Data selection: During the selection process, data relevant to the examination are fetched from the data sets.
4. Data transformation: Data is converted or combined into forms suitable for mining by carrying out summarization or aggregate operations.
5. Data mining: This is a critical cycle where category-wise strategies are useful to extract information strategies.

The following figure displays the KDD process:

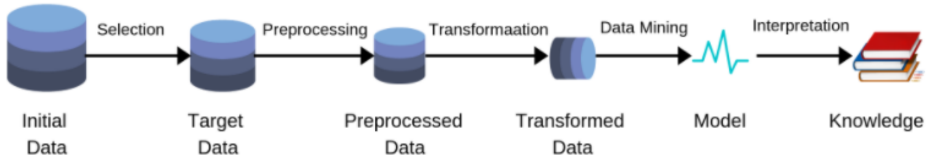

Figure 3:KDD Process

6. Pattern evaluation: This step is used to classify the essentially interesting patterns demonstrating knowledge based on some remarkable measures.
7. Knowledge presentation: Knowledge representation and visualization methods are used to present the mined knowledge to the user.

2.3 Types of Repositories

On a fundamental level, data mining isn't explicit to one sort of media or information. Data mining ought to be material to any sort of data Warehouse. Nonetheless, calculations and approaches may contrast when applied to various kinds of information. For sure, the difficulties introduced by various kinds of information differ altogether. Information mining is being placed into utilization and read for data sets, including social data sets, object-social data sets and article arranged data sets, information stockrooms, conditional data sets, unstructured and semi-organized archives like the World Wide Web, progressed data sets like spatial data sets, interactive media data sets, time-arrangement data sets and literary data sets, and surprisingly level records that are shown in Figure 4:

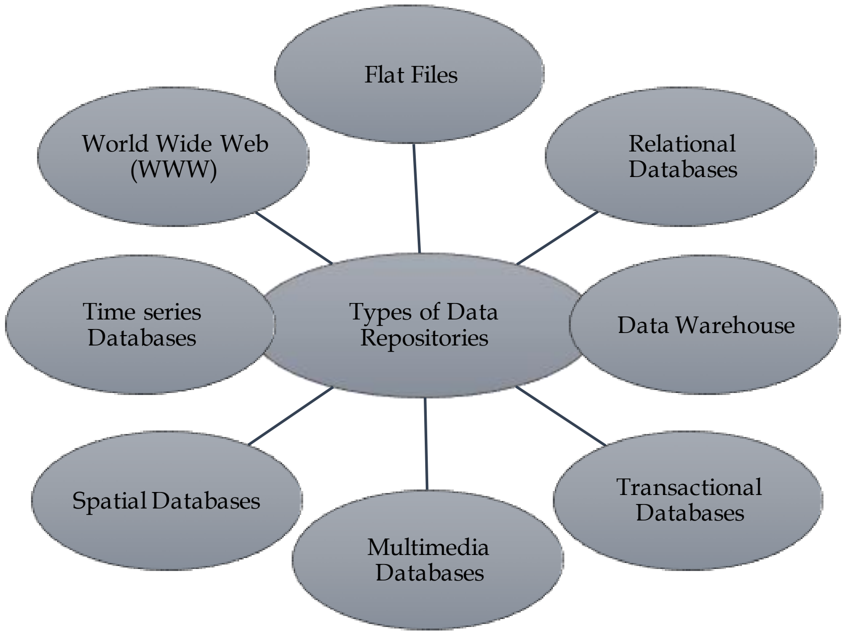

Figure 4: Types of Repositories

Here are some repositories in more detail:

Flat Files: Flat files are well-defined as data collections in text or binary form with an arrangement that can be easily fetched by data mining procedures. Data kept in flat files have no association or pathway between themselves, identically if a traditional database is stored on a flat file, we cannot create any relationship between the tables stored in that database. The data dictionary is used to represent flat files. They are used in a data warehouse to store and conveying information to and from the server, and so forth

Comma Separated Value file.

Relational Databases: Data mining algorithms using traditional databases can be more flexible than other data mining procedures precisely on paper for flat files. Data mining can benefit from SQL for data selection. Operational databases are quite possibly the most ordinarily accessible and most extravagant data archives. Application: Knowledge Extraction, Relational Online Analytical Processing model, etc.

Data Warehouse: A data warehouse center is characterized as the assortment of information incorporated from numerous sources that will inquiries and dynamic and useful for decision making. There are three types of data warehouses: Enterprise data warehouse, Data Mart and Virtual Warehouse. Two approaches can be used to update data in Data Warehouse: Query-driven Approach and Update-driven Approach. A data warehouse is generally demonstrated by a structure that uses multidimensional data that is called a data cube. Each attribute or collection of attributes is represented using dimension in the schema, and each fact table stores the aggregated measures like avg_ sales along with the key values of all the dimension tables. Applications of a data warehouse are commercial decision making, data extraction, etc.

Transactional Databases: A transactional database is a group of data that is prepared by considering different timestamps, dates, etc to characterize transactions in databases. This type of database always ensures consistent data even after updates because in case of any failure the transaction undoes all its updates if some failure occurs until the transaction finally commits its updates. It follows the ACID property of DBMS. Market Basket analysis is an example of typical data-mining analysis with the help of which the retailers they able to identify the frequent patterns by generating associations between the items that are purchased together. Transactional database Applications are banking, distributed systems, Object databases, etc.

Multimedia databases: This database consists of the data of audio, video, and other related files. Such multi-media information we are not able to store in traditional databases so object-oriented databases are used to store such type of information. Multimedia is characterized by high dimensionality, which makes data mining even more challenging. It may require computer vision, computer visuals, image elucidation, and natural language processing. Various applications of multi-media databases are digital libraries, video-on-demand, news-on-demand, musical databases, etc.

Spatial Databases: A spatial database is a database that is optimized for storing and querying data that represents objects defined in a geometric space. Most spatial databases allow the representation of simple geometric objects such as points, lines, and polygons. It stores data in the form of coordinates, topology, lines, polygons, etc. Applications of spatial databases are Maps, Global positioning, etc.

Time series data: A temporal database is a database that has certain features that support time-sensitive status for entries. Where some databases are considered current databases and only support factual data considered valid at the time of use, a temporal database can establish at what times certain entries are accurate. Data mining in time series databases encompasses the study of developments and associations between valuations of different variables as well as a prediction of trends. Various submissions of time series data are stock market data and logging activities.

World Wide Web: WWW states to World wide web is an assembly of different type of documents and resources which includeaudial, visuals, textual, etc which are recognized by Uniform Resource Locators (URLs) with the help of web browsers, connected by HTML pages, and reachable via the network. It is the most mixed and active data repository. Data in the WWW is prepared in interrelated documents. These documents can be audio, video, text, etc. Online shopping, Job search, Research, studying, etc are the various applications of WWW.

2.4 Data Mining Functionalities

Functionalities of data mining are used to identify the kind of patterns to be found in data mining tasks.

Data mining tasks can be categorized as follows:

Descriptive: It comprises certain knowledge to comprehend what is happening within the data without a prior idea. The collective data features are emphasized in the data set.

count, average, etc.

Predictive: It helps developers to offer unlabeled explanations of attributes. Based on earlier tests, the software assesses the absent features.

Judging from the findings of a patient's medical examinations that is he suffering from any particular disease.

Class/Concept Descriptions:

Classes or definitions can be correlated with results. In simplified, descriptive, and yet accurate ways, it can be helpful to define individual groups and concepts. These class or concept definitions are referred to as class/concept descriptions.

Data Characterization:

This refers to the summary of general characteristics or features of the class that is under the study. For example. To study the characteristics of a software product whose sales increased by 15% two years ago, anyone can collect these types of data related to such products by running SQL queries.

Data Discrimination:

It compares common features of class which is under study. The output of this process can be represented in many forms. Eg., bar charts, curves, and pie charts.

Mining Frequent Patterns, Associations, and Correlations:

Patterns that occur frequently in different transactions are termed as frequent patterns. There are several kinds of frequent patterns, including item-sets, and subsequences. A frequent item-set usually refers to a set of items that appear together frequently in a transactional data set, such as pencil and eraser, which are frequently bought together in stationery stores by many customers. The mining of frequent patterns leads to the detection of motivating relations and associations within the data.

Association Analysis

Assuming, as a marketing manager, you would like to define which items are frequently purchased collectively within the alike transactions.

Buys (X, "computer")=buys(X,"software") [support=1%,confidence=50%]

where X is a variable demonstrating a customer. Confidence=50% means that if a customer buys a computer, there is a 50% chance that he or she will purchase software as well. Support =1% means that 1% of all of the transactions under investigation displayed that computers and software were bought together.

Classification and Prediction

Classification is the method of finding a model that describes and differentiates data classes or concepts. The resultant model is based on the exploration of a set of training data for which the class labels are identified. The model is used to forecast the class label of objects for which the class label is unknown. **Regression analysis** is a statistical procedure that is most often used for numeric prediction, though other methods occur as well.

Example: male or female

2.5 Methods of presenting Derived Model

Different methods like classification (IF-THEN) rules, decision trees, mathematical formulae, or neural networks are available to present the derived model. The explanation of each model is as follows:

IF - THEN Rules

The IF part of the rule where we specify the condition is entitled to rule antecedent or precondition. The THEN fragment of the rule is called rule consequent. The antecedent part of the condition comprises of at least one trait test and these tests are consistently AND ed. The subsequent part comprises of the class forecast.

Example R1: (age = youth) ^ (student = yes))(buys computer = yes)

Decision Tree

A decision tree is a structure that represents the data in a hierarchal form that comprises a root node, branches, and leaf nodes. Every non-leaf node signifies a test condition on an attribute, each branch signifies the different outcomes of a test, and each leaf node represents a class label which the classifier is used for prediction. The top node in the tree is the root node. The following decision tree is for the idea buy_ computer that demonstrates whether a client at an organization is probably going to purchase a PC or not. Each interior hub addresses a test on a characteristic. Each leaf hub addresses a class.

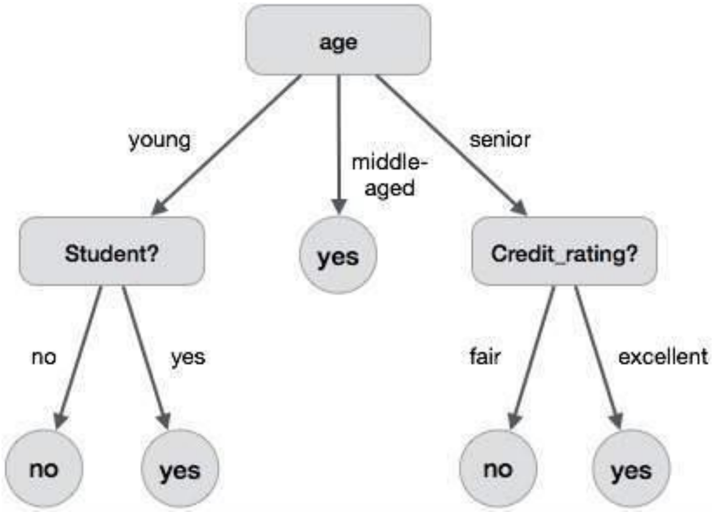

Figure 5: Decision Tree

Cluster Analysis

Unlike classification and prediction, which examine class-labeled data objects, clustering evaluates data objects deprived of referring to a known class label. In general, the class labels do not exist in the training data merely because they are not recognized, to begin with. Clustering can be used to create such labels. The items are clustered or gathered based on the principle of "maximizing the intra-class similarity and minimizing the interclass similarity"

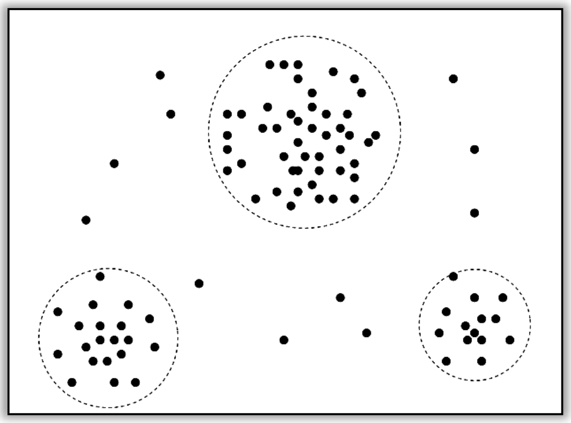

Figure 6: Cluster Analysis

Outlier Analysis

Outlier Analysis is a process that involves identifying the anomalous observation in the dataset. Let us first understand what outliers are. Outliers are nothing but an extreme value that deviates from the other observations in the dataset. Data objects that do not match with the general behavior or model of the data. Most analyses discard outliers as noise or exceptions. Outliers may be detected using statistical tests or using distance measures where objects that are a substantial distance from any other cluster are considered outliers. The examination of outlier data is referred to as outlier mining.

2.6 Data Mining Tasks

Data mining deals with the kind of patterns that can be mined. Based on the kind of data to be mined, there are two categories of tasks involved in Data Mining as shown in Figure 7 −

- Descriptive
- Classification and Prediction

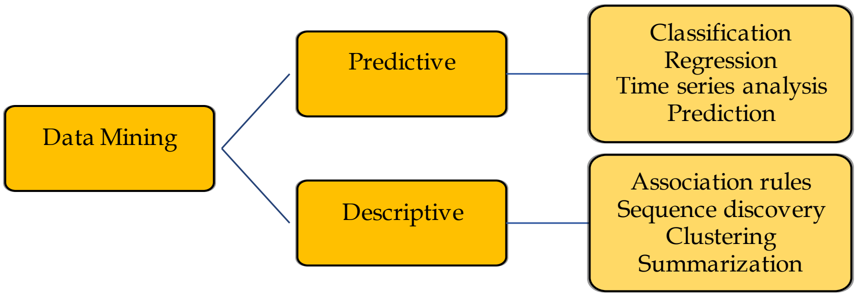

Figure 7: Types of data mining Tasks

Classification

During the learning phase, it constructs a model that classifies the data based on the training set and the values known as class labels and uses these labels for classifying new data.

Data classification is a two-step process:

Learning step (where a classification model is constructed)

Classification step (where the model is used to predict class labels for given data).

In the **learning step** (or training phase), a classification algorithm builds the classifier by analyzing or "learning from" a training set .A tuple, **X**, is represented by an N-dimensional attribute vector,

X ={x1, x2,……..xN}

Each tuple, X, is assumed to belong to a predefined class as determined by another database attribute called the **class label attribute.** The individual tuples making up the training set are referred to as **training tuples** and are randomly sampled from the database under analysis.

pattern recognition

Clustering

Clustering is the unsupervised learning process of building a group the different objects based upon their similarity index. While clustering one thing we need to remember that the cluster quality can be measured as well if and only if the intra-cluster similarity is high and inter-cluster similarity is low. During cluster analysis, we initially partition the data-set into groups based on the data similarity index and then assign the labels to the groups. It is the task of segmenting a dissimilar group into several similar sub-groups or clusters. Similar data items are grouped in one cluster while dissimilar in another cluster.

Bank customer

Time series analysis

Time series analysis comprises methods for analyzing time-series data to extract meaningful statistics and other characteristics of the data. Time series forecasting is the use of a model to predict future values based on previously observed values. The value of an attribute is examined as it varies over time. A time series plot is used to visualize time series.

Stock exchange.

Summarization

Data Summarization is a simple term for a short conclusion of a big theory or a paragraph. This is something where you write the code and, in the end, you declare the final result in the form of summarizing data. Data summarization has great importance in data mining. Abstraction or generalization of data resulting in a smaller set that gives a general overview of data. Alternatively, summary-type information can be derived from data.

Association

Association is a data mining function that discovers the probability of the co-occurrence of items in a collection. The relationships between co-occurring items are expressed as association rules. Association rules are often used to analyze sales transactions. Association is a further prevalent data mining task. Association is also termed as market basket analysis. The goal of the association task is as follows:

- Finding of frequent item-sets
- Finding of association rules.

A frequent item set may look like

{Product = "coke", Product = "Fries", Product = "Squash"}.

Regression

Regression is a data mining function that predicts a number. For example, a regression model could be used to predict the value of a house based on location, number of rooms, lot size, and other factors. A regression task begins with a data set in which the target values are known. The task of regression is related to classification. The key difference is that a probable feature is a continuous number. Regression methods have been extensively studied for centuries on the ground of statistics. Linear regression and logistic regression are the most prevalent regression methods.

2.7 Data Mining Trends

One of the best extensively used methods to fetch data from heterogeneous data sources and establish them for better usage is data mining. Complex algorithms form the source for data mining as they agree for data segmentation to recognize various trends and patterns, detect variations, and predict the chances of various events happening. Some of the key data mining trends are:

- **Application exploration**: The exploration of data mining for businesses continues to expand as e-commerce and e-marketing have become mainstream in the retail industry.
- **Interactive and Scalable data mining methods**: In appearance differently from conventional information examination strategies, data mining should have the option to deal with tremendous measures of information proficiently and, if conceivable, intelligently. Since the measure of information being gathered keeps on expanding quickly, versatile calculations for individual and incorporated data mining capacities become fundamental.
- **Multimedia Data Mining:** It involves the withdrawal of data from different kinds of multimedia devices such as audial, script, hypertext, video, pictures, etc. and the data is transformed into anarithmetic representation in different layouts. This further can be used in clustering and classifications, carrying out similarity checks, and also recognizing associations.
- **Ubiquitous Data Mining:** This technique includes the withdrawal of data from movable devices to get data about different entities. Regardless of having several challenges in this type such as complexity, confidentiality, cost, etc. this method has a lot of prospects to be vast in various trades especially in learning human-computer interactions.
- **Distributed Data Mining**: It involves the withdrawal of an enormous amount of info stored in different business locations or at diverse organizations. Highly refined algorithms are used to fetch data from diverse locations and offer suitable insights and reports based upon the available data.
- **Spatial and Geographic Data Mining:** Type of data mining which includes extracting information from environmental, astronomical, and geographical data which also includes images taken from outer space. This type of data mining can reveal various aspects such as distance and topology which is mainly used in geographic information systems and other navigation applications.
- **Time Series and Sequence Data Mining:** The primary application of this type of data mining is the study of cyclical and seasonal trends. This method is mainly being used by retail companies to access customer's buying patterns and their behaviors.

Data mining trends	Algorithms/ Techniques employed	Data formats	Computing Resources	Prime areas of application
Past	Statistical, machine learning techniques	Numerical data and structured data stored in a traditional database	Evolution of 4G PL and various related techniques	Business
Current	Statistical, machine learning, artificial intelligence, pattern recognition techniques	Heterogeneous data formats include structured, semi-structured, and unstructured data	High-speed networks, high-end storage devices, and parallel distributed computing, etc.	Business, web, medical diagnosis, etc.

Table 1: Past and Current trends in datamining

2.8 Data Mining Issues

Data mining is not an easy task, as the calculations utilized can get exceptionally unpredictable and information isn't generally accessible at one spot. It should be incorporated from different heterogeneous information sources. These variables likewise make a few issues. Here in this instructional exercise, we will examine the significant issues in regards to–

- Mining Methodology and User Interaction
- Performance Issues
- Diverse Data Types Issues

Figure 8 describes the major issues:

Figure 8: Mining Issues

Mining Methodology and User Interaction Issues

- Mining dissimilar kinds of knowledge in databases − Diverse users may be concerned with different types of knowledge. Consequently, it is essential for data mining to cover an extensive range of knowledge discovery tasks.
- Collaborating mining of knowledge at numerous levels of abstraction − The data mining process requests to be communicating because it allows users to emphasize the search for patterns, providing and cleansing data mining requests based on the reverted results.

- Integration of background knowledge − To monitor the discovery process and to prompt the discovered patterns, contextual knowledge can be used. Background knowledge may be used to rapidly the discovered patterns not only in concise terms but at multiple levels of abstraction.
- Data mining query languages and ad hoc data mining − Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- Presentation and visualization of data mining results − Once the patterns are discovered it needs to be expressed in high-level languages, and visual representations. These representations should be easily understandable.
- Handling noisy or incomplete data −Data cleaning methods are a prerequisite to handling the noise and incomplete objects whereas mining the data uniformities. In the absence of data cleaning methods, the accuracy of the discovered patterns will be poor.
- Pattern evaluation −Patterns discovered from the collected data should be motivating because either they signify common knowledge or lack novelty.

Performance Issues

- There can be performance-related issues such as follows −
- Efficiency and scalability of data mining algorithms − To effectively extract the information from a huge amount of data in databases, the data mining algorithm must be efficient and scalable.
- Parallel, distributed, and incremental mining algorithms − The factors such as the huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which are further processed in a parallel fashion. Then the results from the partitions are merged. The incremental algorithms, update databases without mining the data again from scratch.

Diverse Data Types Issues

- Handling of relational and complex types of data − The database may contain complex data objects, multimedia data objects, spatial data, temporal data, etc. One system can't mine all this kind of data.
- Mining information from heterogeneous databases and global information systems − The data is available at different data sources on LAN or WAN. These data sources may be structured, semi-structured, or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

2.9 Ethical Issues

As with many technologies, both positives and negatives lie in the power of data mining. There are, of course, valid arguments to both sides. Here is the positive as well as the negative things about data mining from different perspectives.

Consumers' Point of View

According to the consumers, data mining benefits businesses more than it benefits them. Consumers may benefit from data mining by having companies customized their products and service to fit the consumers' individual needs. However, the consumers' privacy may be lost as a result of data mining.

Data mining is a major way that companies can invade consumers' privacy. Consumers are surprised at how much companies know about their personal lives. For example, companies may know your name, address, birthday, and personal information about your family such as how many children you have. They may also know what medications you take, what kind of music you listen to, and what are your favourite books or movies. The lists go on and on. Consumers are afraid that these companies may misuse their information, or not having enough security to protect their personal information from unauthorized access. For example, the incident about the hackers in the Ford Motor company case illustrated how insufficient companies are at protecting their customers' personal information. Companies are making profits from their customers' personal data, but they do not want to spend a lot amount of money to design a sophisticated security system to protect that data. At least half of Internet users interviewed by Statistical Research, Inc. claimed that they were very concerned about the misuse of credit card information given online, the selling or sharing of personal information by different websites, and cookies that track consumers' Internet activity.

Data mining can also be used to discriminate against a certain group of people in the population. For example, if through data mining, a certain group of people was determined to carry a high risk for a deathly disease (eg. HIV, cancer), then the insurance company may refuse to sell an insurance policy to them based on this information. The insurance company's action is not only unethical but may also have a severe impact on our health care system as well as the individuals involved. If these high-risk people cannot buy insurance, they may die sooner than expected because they cannot afford to go to the doctor as often as they should. Also, the government may have to step in and provide insurance coverage for those people, thus would drive up the health care costs.

Organizations' Point of View

Data mining is a dream that comes true to businesses because data mining helps enhance their overall operations and discover new patterns that may allow companies to better serve their customers. Through data mining, financial and insurance companies can detect patterns of fraudulent credit card usage, identify behaviour patterns of risk customers, and analyze claims. Data mining would help these companies minimize their risk and increase their profits. Since companies can minimize their risk, they may be able to charge the customers lower interest rates or a lower premium. Companies are saying that data mining is beneficial to everyone because some of the benefits that they obtained through data mining will be passed on to the consumers. When it comes to privacy issues, organizations are saying that they are doing everything they can to protect their customers' personal information. Besides, they only use consumer data for ethical purposes such as marketing, detecting credit card fraud, etc. To ensure that personal information is used ethically, the chief information officer (CIO) Magazine has put together a list of what they call the Six Commandments of Ethical Date Management. The six commandments include:

1. Data is a valuable corporate asset and should be managed as such, like cash, facilities, or any other corporate asset;

2. The CIO is a steward of corporate data and is responsible for managing it over its life cycle (from its generation to its appropriate destruction);

3. The CIO is responsible for controlling access to and use of data, as determined by governmental regulation and corporate policy;

4. The CIO is responsible for preventing inappropriate destruction of data;

5. The CIO is responsible for bringing technological knowledge to the development of data management practices and policies;

6. The CIO should partner with executive peers to develop and execute the organization's data management policies.

Government Point of View

The government is in a dilemma when it comes to data mining practices. On one hand, the government wants to have access to people's personal data so that it can tighten the security system and protect the public from terrorists, but on the other hand, the government wants to protect people's privacy rights. The government recognizes the value of data mining to society, thus wanting the businesses to use the consumers' personal information ethically. According to the government, it is against the law for companies and organizations to trade data they had collected for money or data collected by another organization. To protect people's privacy rights, the government wants to create laws to monitor data-mining practices. However, it is extremely difficult to monitor such disparate resources as servers, databases, and websites. Also, the Internet is global, thus creating tremendous difficulty for the government to enforce the laws.

Society's Point of View

Data mining can aid law enforcers in their process of identify criminal suspects and apprehend these criminals. Data mining can help reduce the amount of time and effort that these law enforcers have to spend on any one particular case. Thus, allowing them to deal with more problems. Hopefully, this would make the country becomes a safer place. Also, data mining may also help reduce terrorist acts by allowing government officers to identify and locate potential terrorists early. Thus, preventing another incidence likes the World Trade Center tragedy from occurring on American soil.

Data mining can also benefit society by allowing researchers to collect and analyze data more efficiently. For example, it took researchers more than a decade to complete the Human Genome Project. But with data mining, similar projects could be completed in a shorter amount of time. Data mining may be an important tool that aids researchers in their search for new medications, biological agents, or gene therapy that would cure deadly diseases such as cancers or AIDS.

Summary

- The information and knowledge gained can be used for applications ranging from business management, production control, and market analysis, to engineering design and science exploration.
- Data mining can be viewed as a result of the natural evolution of information technology.
- An evolutionary path has been witnessed in the database industry in the development of data collection and database creation, data management, and data analysis functionalities.
- Data mining refers to extracting or "mining" knowledge from large amounts of data. Some other terms like knowledge mining from data, knowledge extraction, data/pattern analysis, data archaeology, and data dredging are also used for data mining.
- Knowledge discovery is a process and consists of an iterative sequence of data cleaning, data integration, data selection data transformation, data mining, pattern evaluation, and knowledge presentation.

Keywords

Data: Data are any facts, numbers, or text that can be processed by a computer.

Data mining: Data mining is the practice of automatically searching large stores of data to discover patterns and trends that go beyond simple analysis.

Data cleaning: To remove noisy and inconsistent data.

Data integration: Multiple data sources may be combined.

Data selection: Data relevant to the analysis task are retrieved from the database.

Data transformation: Where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.

KDD: Many people treat data mining as a synonym for another popularly used term.

Knowledge presentation: Visualisation and knowledge representation techniques are used to present the mined knowledge to the user.

Pattern evaluation: To identify the truly interesting patterns representing knowledge based on some interestingness measures.

Self Assessment

Choose the appropriate answers:

1. KDD stands for

(a) Knowledge Design Database

(b) Knowledge Discovery Database

(c) Knowledge Discovery Design

(d) Knowledge Design Development

2. What is true about data mining?

A. Data Mining is defined as the procedure of extracting information from huge sets of data
B. Data mining also involves other processes such as Data Cleaning, Data Integration, Data Transformation
C. Data mining is the procedure of mining knowledge from data.
D. All of the above.

3. What Data mining **is?**

a) Time variant non-volatile collection of data

b) The actual discovery phase of a knowledge

c) The stage of selecting the right data

d) None of these

4. ………………………..is not a data mining functionality?

A) Clustering and Analysis

B) Selection and interpretation

C) Classification and regression

D) Characterization and Discrimination

5. …………………..is the output of KDD

a) Query

b) Useful Information

c) Data

d) Information

6. Which of the following is not belong to data mining?

(A). Knowledge extraction

(B). Data transformation

(C). Data exploration

(D). Data archaeology

7.…………………………….. databases include video, images, audio, and text media.

8. Time-series databases contain time-related data such as…………………….

9. A …………………………………… is a set of heuristics and calculations that creates a data mining model from data.
10. ……………………………………. type of algorithm finds correlations between different attributes in a dataset.

Answer for Self Assessment

1.	B	2.	D	3.	B	4.	B	5.	B
6.	B	7.	Multimedia	8.	Stock market data	9.	Data mining algorithm	10.	Association algorithms

Review Questions

1. What is data mining? How does data mining differ from traditional database access?
2. Discuss, in brief, the characterization of data mining algorithms.
3. Briefly explain the various tasks in data mining.
4. Distinguish between the KDD process and data mining.
5. Define data, information, and knowledge.
6. Explain the process of knowledge discovery.
7. Elaborate on the issues of data mining.

Further Readings

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